

DistriNVS: Distributional Geometry Transport and Visibility-Constrained Completion for Cross-Endoscope Novel-View Synthesis

Anonymous Authors

Abstract—Cross-endoscope novel-view synthesis must both transport tissue observed by the source camera and infer tissue revealed only from the target viewpoint. These two regimes are confounded when sparse or erroneous depth is projected as deterministic points: sampling gaps, depth failures, surface collisions, and genuine disocclusions all become visually identical holes. We present DistriNVS, a reliability-aware framework that models every source depth as a distribution and propagates it through calibrated geometry. Its distributional surface splatting renderer produces an anisotropic target-plane footprint together with support, depth variance, and collision entropy. These statistics drive reliability-constrained transport-synthesis fusion: a bounded residual expert refines transported appearance, whereas an uncertainty-conditioned Fourier network predicts unsupported or geometrically unreliable content. Their reliability-weighted RGB-D mixture and an explicit risk head are trained with target-depth and backward-reprojection constraints. The prescribed protocol trains on a fixed set of 13 iMED scenes and evaluates without adaptation on held-out EndoVis datasets 8 and 9 using a strict raw-overlap protocol.

I. INTRODUCTION

Robot-assisted minimally invasive surgery depends on endoscopic vision for scene perception and instrument guidance [1], [2]. Cross-endoscope novel-view synthesis can broaden this visual context by predicting a target-endoscope view from synchronized appearance and geometry acquired by a spatially displaced source endoscope [3]. Unlike left-to-right synthesis within one stereo endoscope, the target may observe anatomy absent from both source cameras. The task therefore combines two epistemically distinct regimes: co-visible anatomy has measured appearance that should be transported through calibrated geometry, whereas target-only anatomy lacks source evidence and must be synthesized. A reliable system should retain source evidence, identify inferred content, and thereby support wider-field robotic perception, retrospective visualization, and surgical simulation [4]. Such uncertainty-aware cross-view completion may also support visual-state construction for surgical robot world models [5].

Achieving this separation is difficult because endoscopic depth and calibration are unreliable around specularities, weak texture, depth discontinuities, instruments, and deforming tissue [6]. Conventional forward warping represents each valid source pixel by one 3-D point; discretized projection leaves sampling gaps, while multiple points may compete for the same target pixel. A binary validity mask then conflates missing depth, rasterization holes, depth-ordering conflicts, and genuine cross-endoscope disocclusions. Treating these

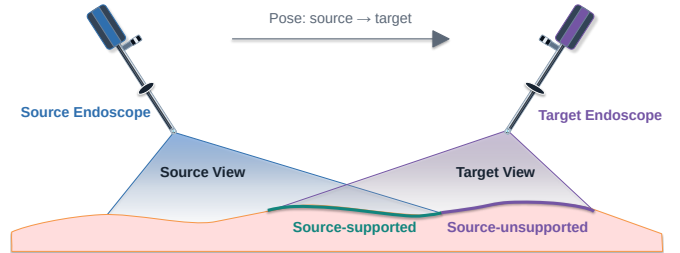


Fig. 1. Cross-endoscope novel-view synthesis under partial field-of-view overlap. Given the calibrated source-to-target pose, co-visible anatomy is source-supported and can be transported geometrically (green), whereas target-only anatomy lacks source evidence and must be synthesized (purple). DistriNVS represents this provenance boundary through reliability-conditioned routing between transported and synthesized RGB-D.

cases uniformly either degrades observed structure or permits hallucination where geometric evidence remains trustworthy.

Existing methods address complementary parts of this problem but do not expose the required evidence boundary. Differentiable point rendering and softmax splatting improve feature transport and collision handling [7], [8], while reference-guided inpainting combines depth-based alignment with learned completion [9]. Large-mask inpainting networks provide strong global image priors [10], [11], and endoscopic video completion adds depth, temporal, or stereo cues [12]–[14]. Endoscopic neural rendering has likewise advanced sparse-view reconstruction and uncertainty modeling [15], [16]. However, deterministic projection commonly assigns equal reliability to all valid depths, while image-space uncertainty does not explicitly describe how source geometry supports a spatially offset target view. Neither approach propagates per-pixel depth uncertainty through the cross-endoscope transform and uses the resulting visibility statistics to constrain where appearance may be synthesized.

Our central premise is to *transport reliable geometric evidence and synthesize only unsupported or geometrically unreliable regions*. We introduce DistriNVS, a reliability-aware framework for direct cross-endoscope novel-view synthesis. Its distributional surface splatting (DSS) represents each source depth as a distribution whose bounded mean and scale, surface normal, and confidence are predicted by a depth reliability encoder. DSS projects deterministic sigma samples through the calibrated cross-endoscope transform and aggregates source features using Jacobian- and normal-aware footprints with soft nearest-surface visibility, yielding target support, depth variance, and collision entropy. These

statistics constrain a soft mixture between bounded residual transport for co-visible regions and uncertainty-conditioned Fourier completion for unsupported or ambiguous regions. Target-depth and risk prediction plus backward RGB-D reprojection constrain this mixture during training.

The paper makes three contributions:

- We formulate cross-endoscope novel-view synthesis as a reliability-aware separation between transport of anatomy observed by the source endoscope and constrained inference of content revealed only by the target endoscope.
- We introduce DSS, a differentiable renderer that propagates depth distributions through calibrated cross-endoscope geometry and exposes target support, depth variance, and collision entropy beyond binary validity.
- We design a two-expert decoder that softly mixes bounded residual transport with uncertainty-conditioned Fourier completion under a renderer-derived reliability constraint.

II. RELATED WORK

We briefly review image completion, geometry-guided view transport, and endoscopic neural rendering, with particular attention to how existing approaches represent missing content, exploit calibrated geometry, and quantify uncertainty.

A. Image and Endoscopic Video Inpainting

Generic image inpainting combines mask-aware feature extraction with increasingly broad appearance priors. Partial and gated convolutions prevent invalid activations from contaminating local features [17], [18], whereas LaMa and UFFC use Fourier operations for global context and MAT restricts attention to valid tokens [10], [11], [19]. Recent diffusion models such as BrushNet and TurboFill incorporate masked-image features into pretrained generators or adapt few-step sampling for efficient completion [20], [21]. Despite strong appearance priors, these methods define holes in a fixed image coordinate system and do not describe geometric provenance. Endoscopic completion adds domain cues: temporal learning removes specularities, DAEVI combines spatio-temporal RGB and depth, and SSIFNet fuses flow and stereo-temporal information for instrument occlusions [12]–[14]. Such methods exploit neighboring frames or views but still restore defects defined in the original camera coordinates. Cross-endoscope synthesis instead creates holes through viewpoint transfer: target-only anatomy is unobserved, while rasterization gaps or depth conflicts may retain source support. DistriNVS therefore uses calibrated projection statistics to distinguish true disocclusions from rasterization gaps and depth conflicts.

B. Geometry-Guided Transport and Completion

Geometry-guided view synthesis preserves correspondence by transporting source evidence before generating missing content. SynSin predicts and projects a feature

point cloud, softmax splatting resolves many-to-one mappings with learned importance, and GeoFill refines depth and pose before reprojection and inpainting [7]–[9]. Recent feed-forward Gaussian renderers use stronger explicit representations: MVSplat localizes Gaussians with plane-sweep cost volumes, whereas DepthSplat combines multi-view correspondence with monocular depth priors [22], [23]. They reduce per-scene optimization but infer a radiance representation from multiple calibrated observations, without an explicit per-pixel boundary between measured and synthesized appearance. Generative methods target wider gaps. LeftRefill frames reference-conditioned synthesis as paired-canvas completion, MultiDiff conditions video diffusion on depth-warped reference views, and point-guided latent diffusion uses stereo-derived clouds for laparoscopic view synthesis [24]–[26]. These models can generate plausible structures in large disocclusions, yet plausibility alone does not reveal which pixels remain geometrically supported. DistriNVS instead transports a calibrated source observation with distributional splatting, derives support, variance, and collision statistics during rasterization, and uses them to constrain soft fusion between transported and synthesized content.

C. Endoscopic Novel-View Synthesis and Uncertainty

Endoscopic neural rendering must handle non-rigid tissue, view-dependent illumination, tool occlusions, and sparse camera coverage. EndoNeRF represents deformation with a neural radiance field, while EndoSurf combines dynamic modeling with a neural surface [27], [28]. EndoSparse introduces geometric and appearance priors into Gaussian splatting, and Endo-GSMT uses depth, dense-displacement, and motion priors for dynamic reconstruction [15], [29]. GastroNVS further shows that large viewpoint changes in real gastroscopy amplify geometric artifacts and inter-view illumination mismatch [30]. These systems build persistent scene representations from multiple frames and principally address reconstruction fidelity within a captured scene. Uncertainty-aware approaches address sparse evidence from another direction: UC-NeRF conditions a radiance field on multi-view uncertainty, whereas ExtraGS uses uncertainty to sample virtual cameras and weight diffusion-refined pseudo-observations for extrapolation [4], [16]. Thus, neither uncertainty nor generative priors are new to endoscopic rendering. The distinction is operational: DSS propagates a distribution for each input depth through the calibrated cross-endoscope transform and derives target support, depth variance, and collision entropy from the same rasterization that transports appearance. DistriNVS uses these quantities to route completion during unseen-sequence inference without target-scene optimization.

III. METHOD

A. Problem Formulation and Overview

After resizing, both endoscope views have spatial size $H \times W$; we omit the batch dimension throughout. DistriNVS receives source RGB $\mathbf{I}_s \in [0, 1]^{3 \times H \times W}$, sparse metric depth

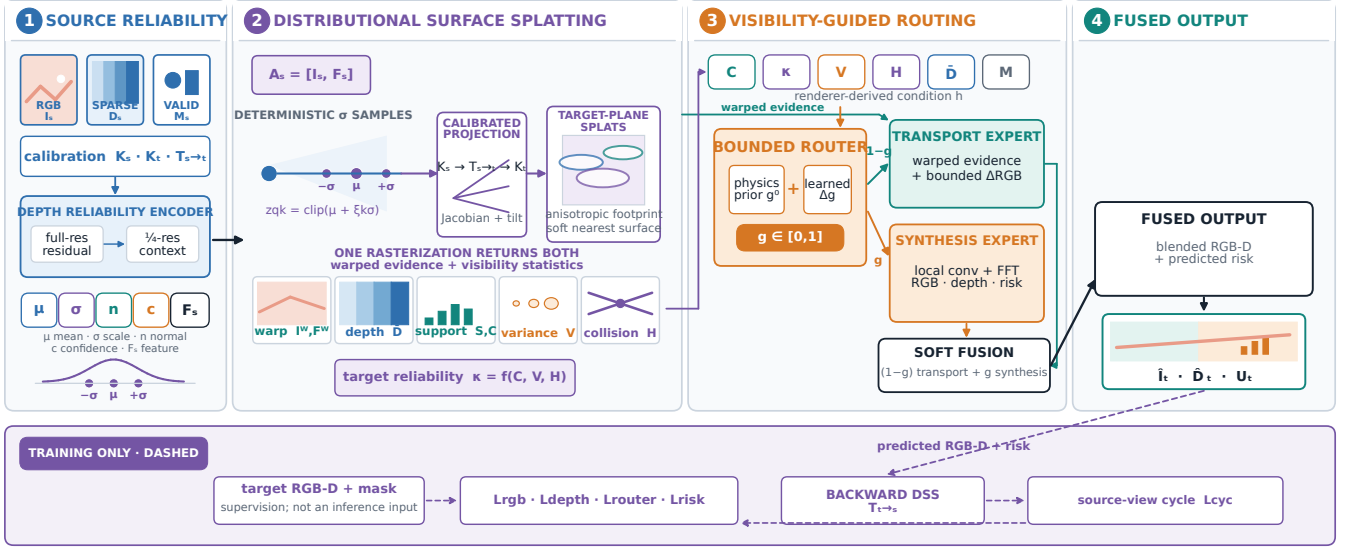


Fig. 2. DistriNVS framework. A reliability encoder predicts source geometry and features. DSS transports deterministic depth hypotheses through the calibrated cross-endoscope transform and jointly produces warped evidence and target-plane visibility statistics. These statistics condition a reliability-constrained soft mixture of a residual transport expert and a Fourier synthesis expert. Solid arrows denote inference; the dashed lower lane is used only during training.

$\mathbf{D}_s \in \mathbb{R}_+^{1 \times H \times W}$, and validity mask $\mathbf{M}_s \in \{0, 1\}^{1 \times H \times W}$. Calibration comprises intrinsics $\mathbf{K}_s, \mathbf{K}_t \in \mathbb{R}^{3 \times 3}$ and a rigid transform $\mathbf{T}_{s \rightarrow t} \in \text{SE}(3) \subset \mathbb{R}^{4 \times 4}$. The model predicts target RGB $\hat{\mathbf{I}}_t \in [0, 1]^{3 \times H \times W}$, depth $\hat{\mathbf{D}}_t \in \mathbb{R}_+^{1 \times H \times W}$, and risk $\mathbf{U}_t \in [0, 1]^{1 \times H \times W}$. Target RGB-D and the target instrument mask are used only for training or evaluation.

Figure 2 summarizes the inference pipeline. A reliability encoder estimates source geometry and uncertainty. Distributional surface splatting (DSS) then transports source evidence and measures target-view support and ambiguity. These measurements route complementary transport and synthesis experts, whose RGB-D predictions are combined by a reliability-constrained soft gate. The backward photometric cycle and risk objective are used only during training.

B. Reliability-Aware Source Geometry

Naive depth completion erases the distinction between measured and inferred geometry. The reliability encoder instead processes source RGB, median-normalized depth, and its validity mask. It produces a feature map $\mathbf{F}_s \in \mathbb{R}^{d_f \times H \times W}$ and scalar fields $\mu, \sigma, c \in \mathbb{R}^{1 \times H \times W}$. They represent depth mean and scale in millimetres, and normalized confidence. It also predicts a unit orientation field $\mathbf{n} \in \mathbb{R}^{3 \times H \times W}$. The normalization scale s_D is a positive scalar in millimetres, chosen as the median valid source depth, and missing inputs are set to zero. During training, $\mathbf{M}_s$ is the post-dropout input mask, whereas $\mathbf{M}_s^{\text{gt}}$ has the same dimensions and retains the pre-dropout supervision.

Measured depths receive a bounded residual correction; missing depths use a positive completion branch. Both obey the physical range, but disjoint confidence intervals keep inferred geometry less authoritative than measurements. The optical-axis-stabilized orientation proxy is learned only through downstream objectives. It serves as a footprint cue

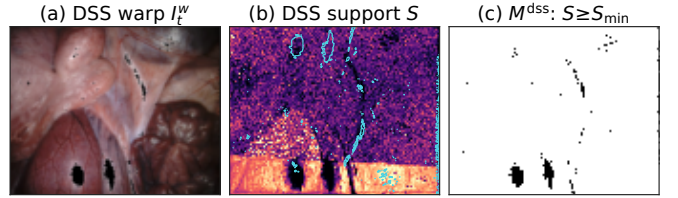


Fig. 3. How continuous DSS support becomes a valid warp, illustrated on a zero-shot EndoVis dataset 8 result. (a) The completed-geometry DSS warp; unsupported pixels are black. (b) Unnormalized support, with colors capped at q_{vis} only for display; the cyan contour marks the validity threshold. (c) The exact render-validity mask is computed from uncapped support.

rather than a metrically supervised normal. Fixed ranges and clipping operations appear in Sec. IV-B.

C. Distributional Surface Splatting

DSS concatenates source RGB and features as $\mathbf{A}_s = [\mathbf{I}_s, \mathbf{F}_s]$ and transports it to a target tensor $\mathbf{A}_t^w$. Both tensors contain $3 + d_f$ channels over the $H \times W$ grid. DSS simultaneously predicts expected depth $\bar{\mathbf{D}}_t \in \mathbb{R}_+^{1 \times H \times W}$. Support S , coverage C , depth variance V , collision entropy $\mathcal{H}$, confidence κ , and render validity M^{dss} are $1 \times H \times W$ target-grid maps. Support and variance are non-negative; coverage, collision entropy, and confidence lie in $[0, 1]$; validity is binary. All arise from one rasterization of a compact depth distribution at each source pixel. Expected depth is in millimetres and variance in square millimetres; the remaining evidence maps are dimensionless.

Depth hypotheses. For each source pixel q , depth hypothesis z_{qk} corresponds to a fixed standardized offset $\xi_k \in \Xi$ and quadrature weight a_k . The set and weights are specified in Sec. IV-B. The calibrated pinhole operator projects each hypothesis to a target coordinate $\mathbf{x}_{qk} \in \mathbb{R}^2$ and target

depth $z_{qk}^t \in \mathbb{R}_+$. It comprises source back-projection, rigid transformation, and target projection [7], [8].

Uncertainty-aware visibility. Depth uncertainty is converted into a hypothesis-dependent, axis-aligned target footprint. The projection derivative J_{qk}^α measures the sensitivity of image coordinate $\alpha \in \{u, v\}$ to source depth. It has units of pixels per millimetre. The pixel-space footprint is

$$r_{qk}^\alpha = b + |J_{qk}^\alpha| \sigma_q + b_n(1 - |n_q^z|). \quad (1)$$

The radii r_{qk}^α , b , and b_n are measured in pixels, while n_q^z is the optical-axis component of the orientation proxy. For target pixel p , ρ_{qkp} denotes its elliptical distance from the projected hypothesis, normalized by both radii. The corresponding contribution is

$$w_{qk \rightarrow p} = c_q a_k \exp \left[-\frac{\rho_{qkp}^2}{2} - \frac{[z_{qk}^t - z_{\min}(p)]_+}{\tau_z} \right]. \quad (2)$$

Source confidence and quadrature mass scale this contribution. Spatial proximity and soft nearest-surface visibility form the exponential penalties. Here, $z_{\min}(p)$ is the detached nearest arrival, $[\cdot]_+$ is the positive part, and τ_z is the visibility temperature in millimetres. Nearest-depth selection and integer assignment are discrete; all remaining operations are differentiable.

Reliability statistics. Let Ω_p collect the retained source-hypothesis contributions to target pixel p . Their total weight defines raw support S_p . At render-valid pixels, DSS normalizes these weights over Ω_p . Their weighted moments yield the transported tensor, expected depth, and conditional depth variance. Coverage is a bounded saturation of raw support. The normalization constant and exact coverage transform are reported in Sec. IV-B.

DSS couples normalized weight entropy to depth dispersion. Let $H_N(p)$ denote the normalized Shannon entropy of the retained weights at pixel p . Collision ambiguity is

$$\mathcal{H}_p = H_N(p) [1 - \exp(-V_p/\tau_z^2)]. \quad (3)$$

This coupling discounts entropy caused only by benign sub-pixel overlap. We derive target confidence κ_p by penalizing coverage with depth variance and collision ambiguity. A pixel is render-valid when its raw support reaches $S_{\min}$, as shown in Fig. 3. Invalid pixels retain support and coverage but zero the transported outputs and ambiguity maps. The continuous evidence therefore distinguishes weak support from conflicting depth order rather than reducing reliability to a binary mask.

D. Reliability-Guided Expert Fusion

Physics-guided routing. Both experts receive transported evidence and a condition tensor $\mathbf{h} \in \mathbb{R}^{6 \times H \times W}$. Its channels are coverage, confidence, normalized variance, collision entropy, normalized depth, and DSS validity. An analytic prior $g^0 \in [0, 1]^{1 \times H \times W}$ favours synthesis as geometric reliability

decreases and equals one outside DSS support:

$$\begin{aligned} \phi_p &= \alpha_\kappa(1 - \kappa_p) + \alpha_V \bar{V}_p + \alpha_H \mathcal{H}_p, \\ g_p^0 &= \begin{cases} \text{clip}_{[0,1]}(\phi_p), & M_p^{\text{dss}} = 1, \\ 1, & M_p^{\text{dss}} = 0. \end{cases} \end{aligned} \quad (4)$$

The provisional synthesis probability is

$$\tilde{g}_p = \text{sigmoid}[\ell_p^0 + \eta_g \tanh R_\theta(\mathbf{h})_p]. \quad (5)$$

Here, $\ell^0 = \text{sg}[\text{logit}(\text{clip}_{[\epsilon_g, 1-\epsilon_g]} g^0)]$, and the residual router R_θ outputs a scalar target-grid map. The factor η_g bounds its correction. A binary routing mask M^{rel} accepts DSS evidence only where the render is valid, $S_p \geq S_{\text{rel}}$, and $V_p \leq V_{\text{rel}}$. The effective synthesis gate is

$$g = (1 - M^{\text{rel}}) \odot \tilde{g}. \quad (6)$$

Routing therefore retains a geometric initialization, suppresses synthesis on reliable DSS support, and adapts continuously elsewhere. The prior coefficients and thresholds are reported in Sec. IV-B. All three maps have shape $1 \times H \times W$.

Complementary experts. The shared expert input has $(9 + d_f) \times H \times W$ channels. The transport expert predicts a bounded RGB correction to the DSS warp, yielding $\mathbf{I}_t^{\text{trans}} \in [0, 1]^{3 \times H \times W}$. It has no depth head, so its depth remains the DSS expectation. The synthesis expert is an uncertainty-conditioned Fourier U-Net [31] that combines local convolution with real-FFT processing. Its outputs ($\mathbf{I}_t^{\text{syn}}, \mathbf{D}_t^{\text{syn}}, \mathbf{U}_t^{\text{syn}}$) have channel dimensions 3, 1, and 1 over the target grid. This local-global design follows Fourier inpainting networks [10], [11], but all weights are trained within DistriNVS. Risk is directly trained as an error-scale proxy on unreliable regions, not as a calibrated failure probability over the full image.

Reliability-constrained output. The transport branch retains source evidence while allowing bounded appearance correction; the synthesis branch supplies content where geometric evidence is weak. Their final predictions are

$$\hat{\mathbf{I}}_t = (1 - g) \odot \mathbf{I}_t^{\text{trans}} + g \odot \mathbf{I}_t^{\text{syn}}, \quad (7)$$

$$\hat{\mathbf{D}}_t = (1 - g) \odot \bar{D}_t + g \odot \mathbf{D}_t^{\text{syn}}, \quad \mathbf{U}_t = \mathbf{U}_t^{\text{syn}}. \quad (8)$$

The gate is broadcast across RGB channels. On M^{rel} , $g = 0$ and the output follows the bounded transport expert rather than an exact copy of the warp; elsewhere the two experts form a soft mixture. For regional supervision, we open the unreliable complement and define $M^{\text{reg}} = M^{\text{dss}} \wedge \neg \text{Open}_k(1 - M^{\text{rel}})$, where M^{dss} is render validity. This cleanup removes isolated unreliable islands from the regional partition; it affects only losses and diagnostics, never the predicted RGB-D or the gate in Eq. (6).

E. Learning Objectives

The training target $\mathbf{I}_t \in [0, 1]^{3 \times H \times W}$ is accompanied by a tissue-valid mask $M_{\text{tis}} \in \{0, 1\}^{1 \times H \times W}$, which excludes annotated instruments when available. The unreliable, reliable, and seam regions are $M_u = (1 - M^{\text{reg}}) \wedge M_{\text{tis}}$, $M_v = M^{\text{reg}} \wedge M_{\text{tis}}$, and $M_b = \text{Band}_w(1 - M^{\text{reg}}) \wedge M_{\text{tis}}$, where Band_w is a width- w dilation-erosion band. Their

Charbonnier terms compare predicted and target RGB; $\mathcal{L}_\nabla$ additionally matches finite-difference image gradients on $M_u \cup M_b$. Training uses this cleaned regional partition, whereas regional image-quality and risk metrics use the raw DSS hole mask.

A masked Laplace negative log-likelihood $\mathcal{L}_{d\text{-nll}}$ supervises the source depth mean and scale on the pre-dropout mask M_s^{gt} . Available target depth D_t shares the dimensions of $\tilde{D}_t$ and supplies a masked relative-depth term $\mathcal{L}_{t\text{-depth}}$. The depth-guided cycle splats the predicted RGB-D back through the inverse calibrated transform. Detached risk controls its depth scale, whereas complementary risk controls its confidence. The cycle term $\mathcal{L}_{\text{cyc}}$ compares backward-rendered RGB with the source image where both are valid. We do not apply a depth-cycle loss.

The detached router target is $y_p = \mathbf{1}\{S_p < S_{\text{rel}} \vee V_p > V_{\text{rel}} \vee \mathcal{H}_p > \mathcal{H}_{\text{rel}}\}$, and $\mathcal{L}_{\text{router}}$ applies binary cross-entropy to the provisional logit in Eq. (5) over the full image. Render-invalid pixels necessarily satisfy the support test because $S_{\text{rel}} > S_{\text{min}}$. For a mask M , $\langle \cdot \rangle_M$ denotes the mean over its selected entries. The scalar e_p is the channel-mean absolute RGB error at pixel p . The positive scale s_p is an affine mapping of predicted risk between fixed bounds. The uncertainty objective is

$$\mathcal{L}_u = \langle e_p/s_p + \log s_p \rangle_{M_u}. \quad (9)$$

Exact risk bounds are given in Sec. IV-B. The target-depth term is omitted when target depth is unavailable. The complete objective is

$$\begin{aligned} \mathcal{L} = & \lambda_h \mathcal{L}_h + \lambda_v \mathcal{L}_v + \lambda_b \mathcal{L}_b + \lambda_\nabla \mathcal{L}_\nabla \\ & + \lambda_d (\mathcal{L}_{d\text{-nll}} + \mathcal{L}_{t\text{-depth}}) + \lambda_c \mathcal{L}_{\text{cyc}} \\ & + \lambda_r \mathcal{L}_{\text{router}} + \lambda_u \mathcal{L}_u. \end{aligned} \quad (10)$$

At inference, the predictor consumes only source RGB-D, camera calibration, and relative pose, and ends at the fused prediction in Eq. (8). Target RGB-D and instrument masks are used only for supervision or evaluation; the backward-cycle branch and deterministic image repair are disabled. The method assumes known zero-skew pinhole intrinsics, a rigid inter-endoscope transform, and an axis-aligned approximation to projected uncertainty; these assumptions delimit its present scope.

IV. EXPERIMENTS

A. Datasets and Zero-Shot Protocol

Training uses the frozen list of 13 iMED Task 2 scene directories supplied in the project configuration [32]. Splits are made at scene level, not at frame level. The training loader uses a synchronized source RGB-depth and target RGB pair at 640×512 resolution. Fifteen percent of valid source depth samples are removed online, with additional 9×9 missing islands, so that the reliability encoder observes both isolated and contiguous depth failures. EndoVis/SCARED datasets 8 and 9 are reserved for final zero-shot evaluation; neither their images nor metrics are used for model selection.

We determine inclusion using a renderer-only pass from the raw valid source depth. A frame is evaluated only if $|M_{\text{raw}}^{\text{dss}}|/(H_I W_I) > 0.5$; the learned depth completion, target image, and target tool mask cannot influence this decision. This reproduces the project’s requested strict overlap criterion and prevents a learned module from selecting favourable frames. The current tables report frame-wise means and sample standard deviations, both rounded to two decimal places. Paired sequence-level confidence intervals and permutation tests will be added once all baseline predictions are frozen.

[TODO: Verify the final dataset provenance, license statement, total training pair count, held-out sequence count, and the immutable list of selected test frames before submission.]

B. Baselines, Metrics, and Implementation

Geometry baselines will include hard point projection, nearest-neighbour small-hole filling, deterministic soft splatting, and softmax splatting. Image completion baselines will include Telea, LaMa, UFFC, MAT, Stable Diffusion 1.5 Inpainting, and the existing LCM+endoscopy-LoRA system. Directly relevant endoscopic baselines (DAEVI and SSIFNet) will be evaluated when code and an equivalent cross-view protocol are available; otherwise their non-comparable reported values will not be inserted into the main table. Every completion method will use the same raw mask, spatial resolution, selected frames, and direct-output evaluation without method-specific image repair.

The existing evaluator reports full-, raw-hole-, raw-visible-, and morphological seam-band PSNR/SSIM, full-image and hole-region LPIPS, known-region drift, risk expected calibration error (ECE), area under the risk-coverage curve (AURC), and a projection-defect taxonomy. The final study will additionally report DISTs, depth AbsRel/RMSE, disparity endpoint error, normal error, latency, throughput, peak memory, and parameter count. Results will be stratified by raw overlap, hole ratio, depth validity, instrument presence, and sequence.

Renderer and reliability. We use $\Xi = (-1, 0, 1)$ and $(a_k) = (0.2741, 0.4518, 0.2741)$; the display percentile in Fig. 3 is $q_{\text{vis}} = 99$ and affects visualization only. Before projection, every sampled depth hypothesis z_{qk} is clipped to $[1, 500]$ mm. Hypotheses affect the radius-one integer neighbourhood, with $\tau_z = 2$ mm and $S_{\text{min}} = 0.03$. Each footprint r_{qk}^α is clipped to $[0.35, 2.5]$ px, with $b = b_n = 0.75$ px. Spatial and final weights no greater than 10^{-5} and 10^{-8} are pruned, and $\epsilon_{\text{dss}} = 10^{-8}$ stabilizes normalization. Coverage uses $C = 1 - \exp(-S)$, and target confidence uses $\kappa = C \exp[-V/(4\tau_z^2) - \mathcal{H}]$. Weight entropy is divided by the logarithm of at least two retained contributions before it is coupled to depth variance.

The reliability encoder fuses full-resolution residual and quarter-resolution context branches to output $d_f = 16$ feature channels. Its fallback median is $s_D = 100$ mm, and valid depth is normalized as $\tilde{D}_s = \text{clip}(D_s/s_D, 0, 5)$ before missing values are set to zero. Measured and completed confidence occupy $[0.5, 1]$ and $[0, 0.35]$, respectively. The

measured-depth correction ratio is bounded by 0.1, whereas the completion branch has a 0.05 positive offset. Depth and uncertainty are clipped to $[1, 500]$ mm and $[0.25, 20]$ mm.

Router and experts. The condition maps use normalized variance $\bar{V} = V/(V + \tau_z^2)$ and normalized depth $\bar{D} = \text{clip}(\bar{D}_t/s_D, 0, 5)/5$. The analytic prior is clipped to $[0, 1]$ and then to $[\epsilon_g, 1 - \epsilon_g]$ before its detached logit ℓ^0 is evaluated. For Eq. (5), the prior weights are $(\alpha_\kappa, \alpha_V, \alpha_H) = (0.65, 0.20, 0.15)$, $\epsilon_g = 10^{-4}$, and $\eta_g = 0.25$; the residual router is zero-initialized. The transport RGB residual is bounded by 0.15. The four-resolution synthesis network uses 3×3 local branches and five output channels; its metric depth is $[\text{softplus}(\cdot) + 0.05]s_D$. The shared base width is 32.

Routing masks and losses. The routing tests use $S_{\text{rel}} = 0.35$, $V_{\text{rel}} = 9 \text{ mm}^2$, and $\mathcal{H}_{\text{rel}} = 1.0$; regional-mask cleanup uses a 7×7 opening kernel. Since collision entropy lies in $[0, 1]$, its configured hard threshold is neutral, although entropy still affects target confidence and soft routing. A separate 7×7 dilation-erosion kernel extracts the training seam. The Charbonnier constant is 10^{-3} , the source-depth likelihood uses a 10^{-3} scale floor, and target relative-depth error uses a 1-mm denominator floor. Equation (9) uses $(s_{\min}, s_{\max}) = (0.01, 0.50)$ in normalized-RGB units. For the backward cycle, the depth scale is $\text{clip}_{[0.25, 20]}[0.25 + 20 \text{sg}(\mathbf{U}_t)]$ mm and its confidence is $\text{clip}_{[0.05, 1]}(1 - \mathbf{U}_t)$.

Optimization. Training is scheduled for 10,000 AdamW steps with learning rate 2×10^{-4} , weight decay 10^{-4} , and three-GPU distributed training. The per-GPU batch size is 1 with four-step gradient accumulation, giving an effective batch size of 12. We use BF16 mixed precision, maximum gradient norm 1, and seed 6666. The implemented loss weights $(\lambda_h, \lambda_v, \lambda_b, \lambda_\nabla, \lambda_d, \lambda_c, \lambda_r, \lambda_u)$ are $(3, 1, 2, 0.2, 0.2, 0.2, 0.1, 0.1)$.

[TODO: Record the final checkpoint, model-selection rule, GPU model, training time, inference warm-up, precision, exact parameter/FLOP counts, and all baseline versions and checkpoints.]

C. Main Comparison

We evaluated every available prediction on the seven-scene iMED validation split. Complete-coverage methods produced predictions for all 1,392 frames, and every included prediction was re-evaluated against the same target image and raw-warp hole mask. The mask covered 25.84% of an image on average. Five Gaussian pipelines covered all seven scenes. The four published reconstruction baselines used dataset-provided ground-truth depth, whereas the additional StereoSurGS pipeline combined SurgicalGS with FoundationStereo-estimated depth [33], [34]. They are reference reconstruction baselines rather than matched feed-forward completion models.

PR-ENDO, EndoGS, and EndoGaussian met the original run-validity criterion on 1/7, 5/7, and 6/7 scenes (199, 994, and 1,193 frames), respectively. We report these method-dependent subsets as diagnostics, but exclude them from ranking and best-value selection. The EndoGaussian diagnostic retains all frames from its six accepted scenes, including

155 frames flagged as degenerate, rather than selectively removing failed frames. D4Recon yielded only degenerate renders on all 1,392 selected frames, while the remaining zero-success runs provided no evaluable render set; assigning either group an image-quality score would therefore be artificial. Table I reports the frame-wise mean and sample standard deviation under this protocol. The DistriNVS row evaluates the primary soft-fusion output directly, without output hard copying or small-component repair.

The primary soft-fusion DistriNVS output achieved the best PSNR and LPIPS among complete-coverage methods. Relative to the strongest image-completion baselines, it improved full-image PSNR by 3.93 dB over LCM and reduced LPIPS by 0.15 relative to LaMa. Inside the raw DSS holes, it improved PSNR by 7.54 dB, SSIM by 0.06, and LPIPS by 0.20 over the strongest corresponding completion baselines. Full-image SSIM was lower than LaMa by 0.05 and Free-SurGS by 0.08, indicating that the gains in pixel fidelity and perceptual distance do not translate into the best global structural score. The partial-coverage diagnostics characterize different subsets and do not support pairwise superiority claims.

We separately evaluated DistriNVS on held-out EndoVis Datasets 8 and 9 using the strict raw overlap > 0.5 inclusion rule. Among 124 candidate frames, 78 met the rule. Table II reports the two datasets separately and jointly.

Across the 78 accepted frames, the archived variant obtained 23.79 ± 2.49 dB PSNR, 0.68 ± 0.12 SSIM, 0.23 ± 0.06 LPIPS, 21.82 ± 3.81 dB hole PSNR, 0.61 ± 0.15 hole SSIM, and 0.28 ± 0.09 hole LPIPS. Baseline inference on Datasets 8 and 9 is not yet complete; consequently, Table II does not support a cross-method zero-shot ranking. Paired confidence intervals and significance tests will be added after the final prediction set is frozen.

D. Ablation Study

Table III reports the nine completed configurations on the same 1,392 frames from the seven-scene iMED validation split. All models were trained for 10,000 steps with seed 6666 and evaluated without small-component repair. Values in the table are frame-wise mean $\pm$ sample SD, and paired 95% confidence intervals below were obtained by resampling the seven scenes. The primary method is the soft fusion (SF) configuration in Eq. (8). The seven completed component variants retain legacy hard composition (HC) and are therefore compared only with the HC reference, rather than treated as removals from the primary model. For compact ablation labels, we use depth completion (DC), distributional depth (DD), anisotropic footprints (AF), physics-guided routing (PGR), RGB-D cycle consistency (RCC), uncertainty loss (UL), and trust-mask cleanup (TMC).

The primary soft-fusion configuration achieved 22.949 dB full-image PSNR, 0.589 SSIM, 27.951 dB hole PSNR, 0.803 hole SSIM, and 24.563 dB seam PSNR. Relative to the legacy hard-copy reference, the paired differences were +2.202 dB for full-image PSNR (95% CI, +1.649 to +2.799) and +2.504 dB for seam PSNR (95% CI, +1.512

TABLE I

COMPARISON ON THE SEVEN-SCENE iMED VALIDATION SPLIT. ROWS WITH $N = 1,392$ COVER ALL SEVEN SCENES; PARTIAL ROWS SUMMARIZE ONLY THEIR ACCEPTED SCENES AND ARE DIAGNOSTIC. EVERY INCLUDED PREDICTION USES THE SAME TARGET AND RAW-WARP HOLE MASK. VALUES ARE FRAME-WISE MEAN $\pm$ SAMPLE SD. BOLD DENOTES THE BEST MEAN AMONG COMPLETE-COVERAGE ROWS ONLY. FS DENOTES FOUNDATIONSTEREO-ESTIMATED DEPTH; THE OTHER GAUSSIAN RUNS USE DATASET GROUND-TRUTH DEPTH. DISTRiNVS USES DIRECT SOFT FUSION WITHOUT OUTPUT HARD COPYING OR SMALL-COMPONENT REPAIR.

Method	N	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	H-PSNR $\uparrow$	H-SSIM $\uparrow$	H-LPIPS $\downarrow$
<i>Gaussian reconstruction—complete coverage</i>							
Deform3DGS [35]	1,392	15.55 $\pm$ 1.78	0.55 $\pm$ 0.04	0.63 $\pm$ 0.06	14.11 $\pm$ 1.37	0.47 $\pm$ 0.10	0.69 $\pm$ 0.06
Endo-4DGS [36]	1,392	16.44 $\pm$ 1.71	0.58 $\pm$ 0.04	0.61 $\pm$ 0.06	15.61 $\pm$ 1.47	0.57 $\pm$ 0.13	0.66 $\pm$ 0.06
Free-SurGS [37]	1,392	16.83 $\pm$ 2.11	0.67 $\pm$ 0.06	0.63 $\pm$ 0.04	17.05 $\pm$ 2.38	0.72 $\pm$ 0.04	0.61 $\pm$ 0.04
SurgicalGS [33]	1,392	17.41 $\pm$ 1.99	0.60 $\pm$ 0.06	0.60 $\pm$ 0.05	16.98 $\pm$ 2.55	0.61 $\pm$ 0.14	0.65 $\pm$ 0.04
StereoSurGS (FS)	1,392	16.37 $\pm$ 1.46	0.52 $\pm$ 0.08	0.59 $\pm$ 0.05	13.59 $\pm$ 1.27	0.39 $\pm$ 0.13	0.70 $\pm$ 0.06
<i>Gaussian reconstruction—partial coverage (diagnostic)</i>							
PR-ENDO [38]	199	15.51 $\pm$ 0.76	0.50 $\pm$ 0.01	0.62 $\pm$ 0.01	12.98 $\pm$ 0.46	0.20 $\pm$ 0.05	0.71 $\pm$ 0.02
EndoGS [39]	994	15.86 $\pm$ 2.19	0.61 $\pm$ 0.06	0.71 $\pm$ 0.11	15.50 $\pm$ 1.97	0.65 $\pm$ 0.08	0.70 $\pm$ 0.07
EndoGaussian [40]	1,193	15.21 $\pm$ 3.56	0.57 $\pm$ 0.26	0.81 $\pm$ 0.10	16.00 $\pm$ 3.15	0.62 $\pm$ 0.28	0.72 $\pm$ 0.08
<i>Image completion</i>							
SD1.5 Inpainting	1,392	16.84 $\pm$ 2.73	0.57 $\pm$ 0.05	0.52 $\pm$ 0.06	14.56 $\pm$ 3.14	0.50 $\pm$ 0.10	0.64 $\pm$ 0.07
LaMa	1,392	18.99 $\pm$ 1.58	0.64 $\pm$ 0.07	0.47 $\pm$ 0.03	20.41 $\pm$ 2.06	0.74 $\pm$ 0.05	0.48 $\pm$ 0.04
LCM + endoscopy LoRA	1,392	19.02 $\pm$ 1.64	0.63 $\pm$ 0.09	0.49 $\pm$ 0.03	19.54 $\pm$ 2.28	0.74 $\pm$ 0.09	0.49 $\pm$ 0.03
DistrINVS (soft fusion; primary)	1,392	22.95 $\pm$ 2.33	0.59 $\pm$ 0.11	0.32 $\pm$ 0.05	27.95 $\pm$ 2.91	0.80 $\pm$ 0.07	0.28 $\pm$ 0.06

TABLE II

LEGACY DISTRiNVS ZERO-SHOT DIAGNOSTIC ON HELD-OUT ENDOVIS DATASETS 8 AND 9. VALUES ARE FRAME-WISE MEAN $\pm$ SD OVER FRAMES SATISFYING raw overlap > 0.5 . THIS ARCHIVED RUN USED OUTPUT HARD COPYING AND SMALL-COMPONENT REPAIR AND WILL BE REPLACED BY DIRECT-OUTPUT RESULTS.

Dataset	Frames	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	H-PSNR $\uparrow$	H-SSIM $\uparrow$	H-LPIPS $\downarrow$
EndoVis 8	48	24.85 $\pm$ 2.00	0.72 $\pm$ 0.12	0.21 $\pm$ 0.04	23.71 $\pm$ 2.74	0.70 $\pm$ 0.09	0.23 $\pm$ 0.06
EndoVis 9	30	22.10 $\pm$ 2.29	0.63 $\pm$ 0.09	0.27 $\pm$ 0.06	18.80 $\pm$ 3.32	0.45 $\pm$ 0.08	0.35 $\pm$ 0.08
Combined	78	23.79 $\pm$ 2.49	0.68 $\pm$ 0.12	0.23 $\pm$ 0.06	21.82 $\pm$ 3.81	0.61 $\pm$ 0.15	0.28 $\pm$ 0.09

TABLE III

SINGLE-SEED ABLATION ON THE SEVEN-SCENE iMED VALIDATION SPLIT ($N = 1,392$ FRAMES). H-PSNR AND H-SSIM USE EACH CONFIGURATION’S RAW DSS HOLE MASK; DRIFT IS THE MEAN ABSOLUTE RGB DEVIATION FROM THE DSS WARP ON THE CLEANED RELIABLE REGION. VALUES ARE FRAME-WISE MEAN $\pm$ SAMPLE SD FROM DIRECT OUTPUTS WITHOUT SMALL-COMPONENT REPAIR. BOLD MARKS THE BEST UNROUNDED IMAGE-QUALITY OR RISK MEAN; HARD-COPY VARIANTS HAVE ZERO DRIFT BY CONSTRUCTION.

Variant	PSNR $\uparrow$	SSIM $\uparrow$	H-PSNR $\uparrow$	H-SSIM $\uparrow$	Seam PSNR $\uparrow$	Drift $\downarrow$	AURC $\downarrow$
<i>Primary output rule and legacy reference</i>							
DistrINVS (SF; primary)	22.95 $\pm$ 2.33	0.59 $\pm$ 0.11	27.95 $\pm$ 2.91	0.80 $\pm$ 0.07	24.56 $\pm$ 2.33	0.05 $\pm$ 0.01	0.02 $\pm$ 0.01
DistrINVS (HC; reference)	20.75 $\pm$ 1.90	0.49 $\pm$ 0.13	27.56 $\pm$ 2.78	0.78 $\pm$ 0.07	22.06 $\pm$ 1.70	0.00 $\pm$ 0.00	0.03 $\pm$ 0.01
<i>Component variants under legacy hard-copy output</i>							
w/o DC	20.68 $\pm$ 1.89	0.49 $\pm$ 0.13	27.76 $\pm$ 2.48	0.79 $\pm$ 0.06	22.11 $\pm$ 1.66	0.00 $\pm$ 0.00	0.03 $\pm$ 0.01
w/o DD ($K = 1$)	20.67 $\pm$ 1.91	0.48 $\pm$ 0.13	27.19 $\pm$ 2.74	0.77 $\pm$ 0.07	21.51 $\pm$ 1.66	0.00 $\pm$ 0.00	0.03 $\pm$ 0.01
w/o AF	20.65 $\pm$ 1.92	0.48 $\pm$ 0.13	27.73 $\pm$ 2.71	0.78 $\pm$ 0.06	22.11 $\pm$ 1.73	0.00 $\pm$ 0.00	0.02 $\pm$ 0.01
w/o PGR	20.75 $\pm$ 1.89	0.49 $\pm$ 0.13	27.41 $\pm$ 2.72	0.77 $\pm$ 0.07	21.82 $\pm$ 1.68	0.00 $\pm$ 0.00	0.03 $\pm$ 0.01
w/o RCC	20.72 $\pm$ 1.91	0.49 $\pm$ 0.13	27.41 $\pm$ 2.73	0.78 $\pm$ 0.07	21.98 $\pm$ 1.70	0.00 $\pm$ 0.00	0.03 $\pm$ 0.01
w/o UL	20.70 $\pm$ 1.89	0.48 $\pm$ 0.13	27.02 $\pm$ 2.60	0.77 $\pm$ 0.07	21.96 $\pm$ 1.71	0.00 $\pm$ 0.00	0.04 $\pm$ 0.01
w/o TMC	20.77 $\pm$ 1.91	0.49 $\pm$ 0.13	27.56 $\pm$ 2.73	0.78 $\pm$ 0.07	22.39 $\pm$ 1.69	0.00 $\pm$ 0.00	0.03 $\pm$ 0.01

to +3.612). AURC decreased from 0.0269 to 0.0206, while reliable-region drift increased from zero to 0.047 (paired 95% CI, +0.042 to +0.052) in normalized RGB units. This quality–fidelity trade-off supports soft fusion as the primary output rule.

Within the legacy hard-copy family, replacing distributional depth with a single deterministic sample produced the clearest consistent degradation: full-image, hole, and seam PSNR decreased by 0.075 dB (95% CI, -0.107 to -0.048), 0.367 dB (95% CI, -0.637 to -0.177), and

0.546 dB (95% CI, -0.801 to -0.311), respectively. Removing the physics router left full-image PSNR nearly unchanged (-0.003 dB; 95% CI, -0.012 to $+0.007$) but reduced seam PSNR by 0.241 dB (95% CI, -0.319 to -0.158). Depth completion and anisotropic footprints showed mixed regional effects: their removal reduced full-image PSNR by 0.069 and 0.095 dB, respectively, but changed hole PSNR by $+0.203$ dB (95% CI, -0.109 to $+0.525$) and $+0.170$ dB (95% CI, $+0.056$ to $+0.313$). The deterministic renderer also reduced mean raw overlap from 0.765 to 0.758, so its hole and seam differences include a modest change in the evaluated masks.

Disabling trust-mask cleanup left hole PSNR and hole SSIM effectively unchanged, while increasing seam PSNR by 0.329 dB (95% CI, $+0.206$ to $+0.458$) and reducing AURC from 0.0269 to 0.0254. The full-image PSNR change was small and inconclusive ($+0.019$ dB; 95% CI, -0.005 to $+0.044$), whereas SSIM decreased by 0.0013 (95% CI, -0.0018 to -0.0008). Cleanup increased the trusted hard-copy ratio from 0.755 to 0.762, but did not improve aggregate seam fidelity under this single-seed evaluation.

Removing the RGB-D cycle caused a small but consistent reduction in full-image PSNR (-0.028 dB; 95% CI, -0.045 to -0.008) and seam PSNR (-0.083 dB; 95% CI, -0.122 to -0.039). Its hole-PSNR change was less conclusive (-0.148 dB; 95% CI, -0.364 to $+0.061$), indicating that the cycle objective primarily provided a modest regularization benefit. Removing the uncertainty loss reduced hole PSNR by 0.541 dB (95% CI, -0.857 to -0.255) and seam PSNR by 0.102 dB (95% CI, -0.171 to -0.038). More importantly, risk AURC increased from 0.0269 to 0.0365, while risk ECE increased from 0.0236 to 0.9421. Thus, within the legacy hard-copy family, the uncertainty objective was essential for calibrated risk estimates and also improved reconstruction quality in the raw DSS holes.

E. Qualitative, Failure, and Efficiency Analysis

Efficiency is central to the ICRA framing. We will measure end-to-end latency including depth reliability estimation, DSS, routing, and both experts; DSS-only timing will also be reported to isolate the renderer cost. Failure analysis will cover specular depth loss, calibration perturbation, thin instruments, large disocclusion, weak texture, and dynamic tissue deformation.

[TODO: Insert the speed/memory/parameter table, calibration-perturbation curves, and blinded expert assessment if available. Do not claim real-time performance until measured on the stated hardware and resolution.]

V. DISCUSSION

The ablation separates two complementary design effects. Within the legacy hard-copy family, distributional depth consistently improved full-image, hole, and seam fidelity, whereas the physics router primarily improved seam quality. Depth completion and the anisotropic footprint produced smaller, region- dependent changes. Trust-mask cleanup slightly increased global SSIM and the trusted hard-copy

ratio, whereas disabling it improved seam PSNR and AURC. The output rule yielded the largest difference: the separately trained primary soft-fusion configuration improved full-image and seam PSNR by 2.20 and 2.50 dB and reduced AURC from 0.0269 to 0.0206 relative to the legacy hard-copy reference, at the cost of 0.047 reliable-region drift. This pattern is consistent with hard copying carrying residual transport error to the output, whereas soft fusion permits bounded appearance correction. Because the two output configurations differ during both training and inference, the comparison does not isolate these two effects. The suite also contains one seed, and its component variants were trained under the legacy output rule; their interactions with the primary soft-fusion configuration are therefore not yet established. Moreover, the zero-shot table still uses the old hard-copy/repair protocol; cross-dataset conclusions must be refreshed from direct no-post outputs before submission.

The method does not remove the fundamental ambiguity of unseen tissue. A low risk score indicates model confidence under the training distribution; it does not certify anatomical truth. Moreover, the current formulation assumes known intrinsics and a rigid inter-camera transform for each pair, processes frames independently, and uses an axis-aligned rather than full-covariance footprint. The archived zero-shot run also varied substantially between the two held-out datasets. Because that run used the legacy output protocol and baseline coverage remains incomplete, it cannot establish uniform cross-dataset generalization or a fair ranking on Datasets 8 and 9. Large calibration errors, non-rigid motion between nominally synchronized frames, or out-of-distribution anatomy may therefore invalidate the transport statistics. Generated output should remain visibly marked and should not be used as a substitute for a live physical endoscopic view.

VI. CONCLUSION

This paper introduces a reliability-aware formulation of cross-endoscope novel-view synthesis and implements it through distributional surface splatting, reliability-constrained transport/synthesis fusion, and joint appearance-geometry risk modeling. The design turns projection support, depth variance, and collision entropy into operational constraints on what the network may synthesize. No-post validation on the seven-scene iMED split showed that the primary soft-fusion configuration improved full-image and seam PSNR by 2.20 and 2.50 dB over the legacy hard-copy reference. Within the legacy component study, distributional depth provided the most consistent image-fidelity benefit, while physics routing primarily improved transition boundaries. These results support direct reliability- constrained fusion as the primary output formulation. The current evidence is limited to one seed, with component variants evaluated under the legacy hard-copy rule; matched multi-seed and held-out zero-shot experiments remain necessary for broader comparative claims.

[TODO: Replace this placeholder with synchronized columns: source RGB, raw hard warp, raw validity, LaMa, LCM, DistriNVS, target RGB, support, variance, collision entropy, and risk. Include at least one success and one failure for each overlap bin.]

Fig. 4. Planned qualitative comparison and failure visualization. Crops and color scales must be identical across methods; generated and source-backed regions will be explicitly marked.

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